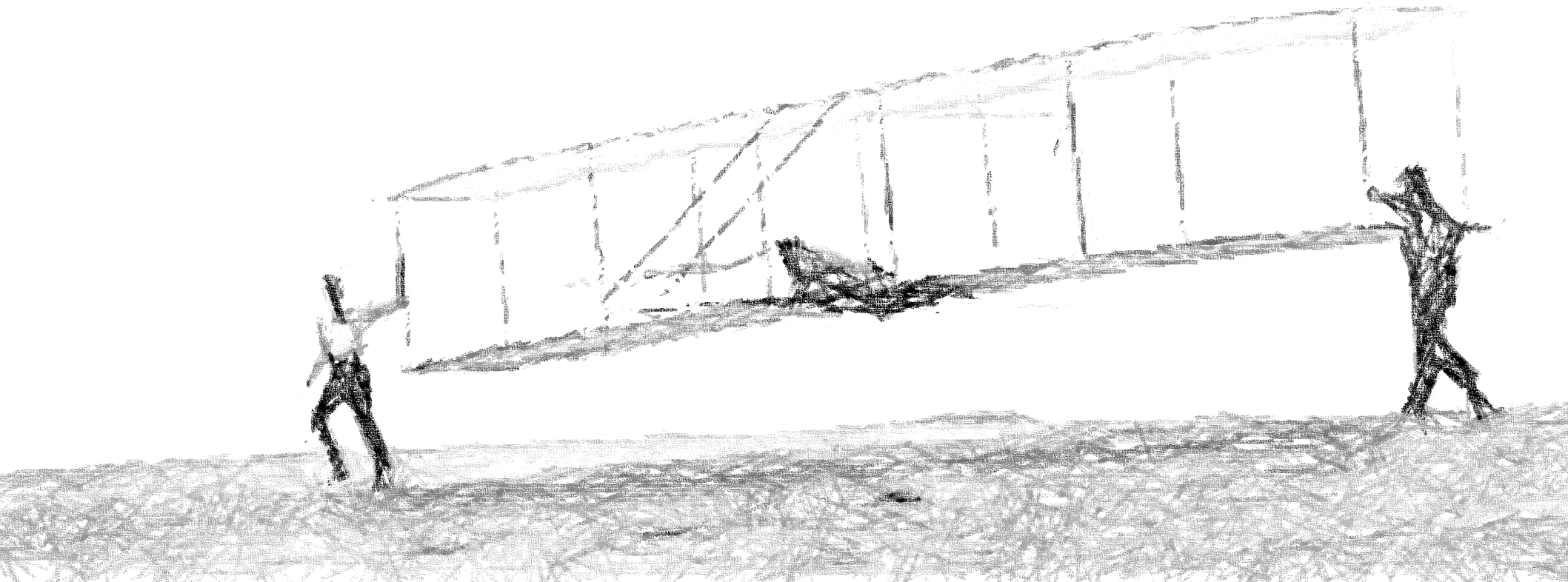




Mechanics applied to Aerospace Engineering

Course 251-14165 – BSc Aerospace Engineering

3. Constraints and reaction forces



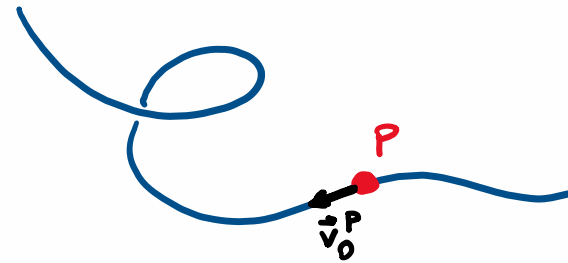
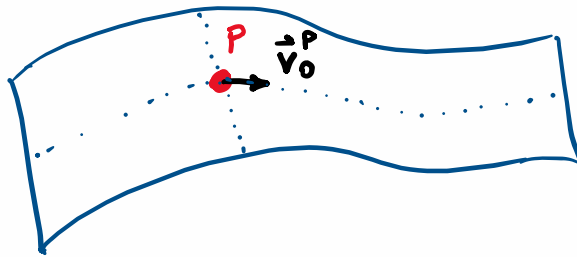
Contents

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Constraints

- In many situations, the motion of a particle is not free, but restricted (*constrained*) by e.g. surfaces or curves. Examples:
 - A ground vehicle that must stay on the surface of a spherical Earth
 - A bead that must slide along a fixed wire of known shape



- We shall consider first *simple constraints* on a single point particle:
 - They can be expressed mathematically as equations on the coordinates of a coordinate system, e.g. x, y, z
 - They do not depend on time

Simple constraints on a single particle are stationary surfaces (two-dimensional spaces) and stationary curves (one-dimensional spaces)

Configuration degrees of freedom

- Simple constraints reduce the number of configuration degrees of freedom (CDOFs) of the particle.

- A free particle in three-dimensional space has 3 CDOFs.
- A free particle in two-dimensional space or constrained to move on a surface has 2 CDOFs.
- A free particle in one-dimensional space or constrained to move on a curve has 1 CDOFs.

Observe that a curve can also be given as the intersection of two surfaces

- There are basically two ways to deal with a simple constraint:
 - We can “ignore it” in our choice of coordinates and use a three-dimensional coordinate system. Then, we add the constraint equations explicitly to our formulation of our problem
 - This is usually not advised, as the differential equations of motion need to be complemented with the constraint equations. Solution of this differential-algebraic system is generally complex
 - We pick a number of generalized coordinates equal to the number of CDOFs of the problem, which *parametrize* the constraint. We do not need to worry about the equations again.
 - This is the path that we will take in this course!

Surfaces and curves

- In Cartesian coordinates, a surface in *implicit form* can be given as: $f(x, y, z) = 0$

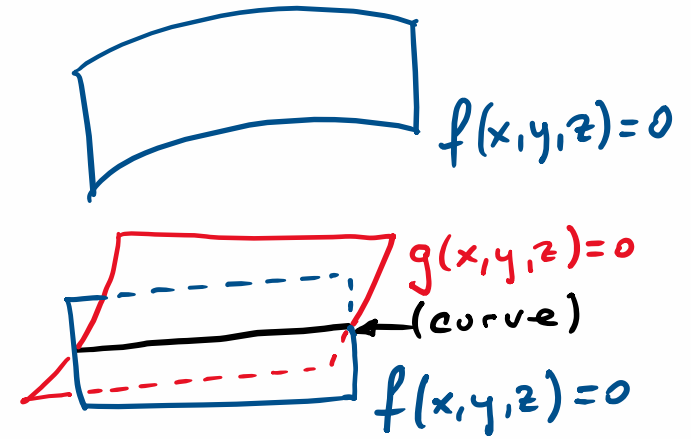
- A curve in implicit form can be given as the intersection of two surfaces:

$$f(x, y, z) = 0; \quad g(x, y, z) = 0$$

- Alternatively, we can express a curve in *parametric form*, in terms of one parameter α (which can be e.g. an arc length parameter)

- Similarly, a surface can also be given in parametric form. However, since it is a two-dimensional object, we need two parameters:

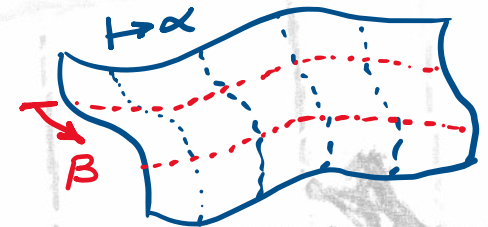
If we have a parametric form of a simple constraint, we have a natural choice of generalized coordinates for the motion of the particle:
 α (in the curve case) or α, β (in the surface case).



$$\begin{cases} x = x(\alpha) \\ y = y(\alpha) \\ z = z(\alpha) \end{cases}$$



$$\begin{cases} x = x(\alpha, \beta) \\ y = y(\alpha, \beta) \\ z = z(\alpha, \beta) \end{cases}$$



It is useful to be able to obtain the implicit form of a surface from the parametric form, and vice-versa.

Examples

- A curve with ellipse shape is given in parametric form. Find an implicit form for the curve:

$$\begin{cases} x = a \cos \alpha \\ y = b \sin \alpha \\ z = 0 \end{cases}$$

- A sphere is given in parametric form. Find an implicit form

$$\begin{cases} x = a \cos \alpha \cos \beta \\ y = a \sin \alpha \cos \beta \\ z = a \sin \beta \end{cases}$$

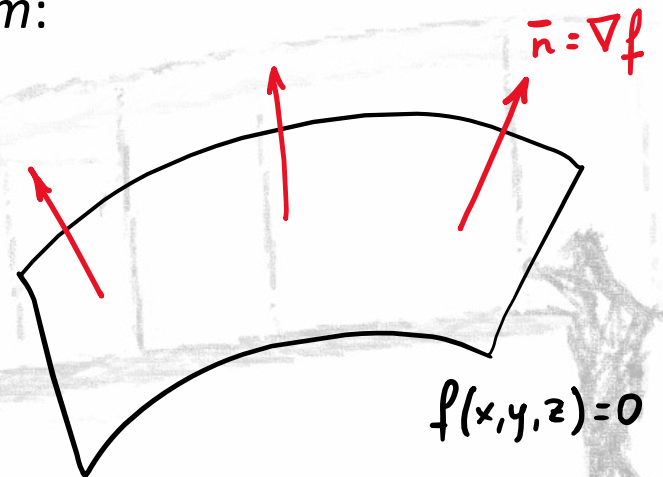
Normal and tangent subspaces: surfaces

- At each point of a smooth surface (two-dimensional manifold) we define
 - A one-dimensional *normal vector subspace* V_N . The vectors of this subspace point in the direction perpendicular to the surface at that point
 - A two-dimensional *tangent vector subspace* V_T , orthogonal to V_N . The vectors in this subspace define the tangent plane to the surface at that point
- The direct sum of these subspaces is equal to the three-dimensional space, $V_N \oplus V_T = \mathbb{R}^3$; $V_N \cap V_T = 0$
- To obtain a normal vector $\mathbf{n} \in V_N$, we can take the *gradient of the surface in implicit form*:
$$f(x, y, z) = 0 \Rightarrow \mathbf{n} = \nabla f$$
 - In general, this vector is not unitary (but can be normalized if necessary)

In Cartesian Coordinates:

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}.$$

The gradient points in the direction in which $f(x, y, z)$ grows in space.

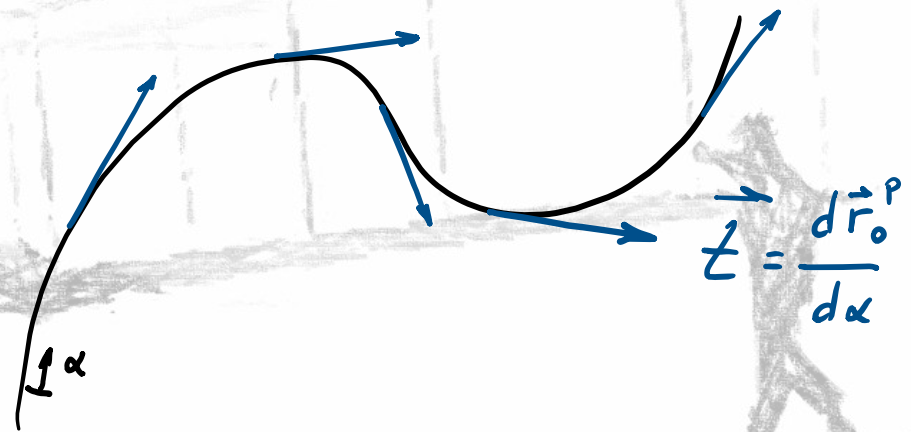


Normal and tangent subspaces: curves

- At each point of a curve (one-dimensional manifold) we define
 - A two-dimensional *normal vector subspace* V_N . The vectors of this subspace point in the directions perpendicular to the curve at that point
 - A one-dimensional *tangent vector subspace* V_T , orthogonal to V_N . The vectors in this subspace are tangent to the curve at that point
- The direct sum of these subspaces is equal to the three-dimensional space, $V_N \oplus V_T = \mathbb{R}^3$; $V_N \cap V_T = 0$
- To obtain a tangent vector $\mathbf{t} \in V_T$, we can differentiate with respect to α in the parametric form of the curve:

$$\begin{cases} x = x(\alpha) \\ y = y(\alpha) \\ z = z(\alpha) \end{cases} \Rightarrow \mathbf{t} = \frac{dx}{d\alpha} \mathbf{i} + \frac{dy}{d\alpha} \mathbf{j} + \frac{dz}{d\alpha} \mathbf{k}$$

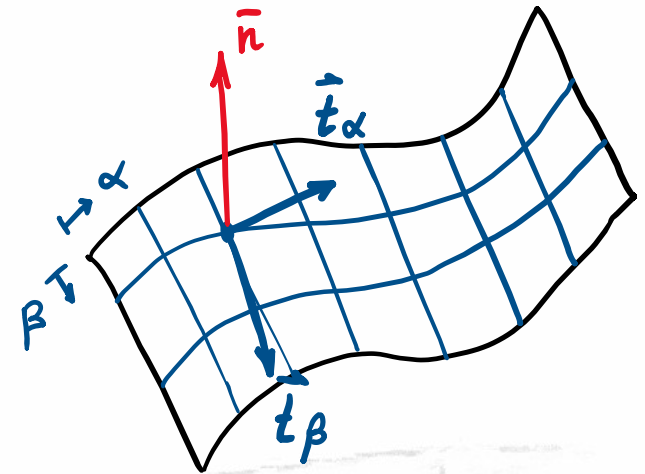
- Observe that $\mathbf{t}$ is not unitary in general (but can be normalized)
 - If α were an arc-length coordinate, $\mathbf{t}$ would be unitary ($\mathbf{e}_t$)



Normal and tangent subspaces: surfaces

- To obtain two independent tangent vectors $\mathbf{t}_\alpha, \mathbf{t}_\beta \in V_T$, we can take partial derivatives with respect to α, β in the surface's parametric form:

$$\begin{cases} x = x(\alpha, \beta) \\ y = y(\alpha, \beta) \\ z = z(\alpha, \beta) \end{cases} \Rightarrow \begin{aligned} \mathbf{t}_\alpha &= \frac{\partial x}{\partial \alpha} \mathbf{i} + \frac{\partial y}{\partial \alpha} \mathbf{j} + \frac{\partial z}{\partial \alpha} \mathbf{k} \\ \mathbf{t}_\beta &= \frac{\partial x}{\partial \beta} \mathbf{i} + \frac{\partial y}{\partial \beta} \mathbf{j} + \frac{\partial z}{\partial \beta} \mathbf{k} \end{aligned}$$



- Or, we may just look for two independent vectors that satisfy $\mathbf{t}_\alpha, \mathbf{t}_\beta \perp \mathbf{n}$
 - Observe that $\mathbf{t}_\alpha, \mathbf{t}_\beta$ are not unitary nor orthogonal in general
- Observe that if we already have $\mathbf{t}_\alpha, \mathbf{t}_\beta$, it is easy to find a valid normal vector $\mathbf{n}$ by doing:

$$\mathbf{n} = \mathbf{t}_\alpha \times \mathbf{t}_\beta$$

- $B_P = \{\mathbf{t}_\alpha, \mathbf{t}_\beta, \mathbf{n}\}$ is a local vector basis at point P on the surface. Observe that this basis is generally not unitary, orthogonal, nor right-handed.

Normal and tangent subspaces: curves

- To obtain two normal vectors $\mathbf{n}_f, \mathbf{n}_g \in V_N$, we can take the gradient of each equation in the implicit form of the curve:

$$f(x, y, z) = 0 \Rightarrow \mathbf{n}_f = \nabla f$$

$$g(x, y, z) = 0 \Rightarrow \mathbf{n}_g = \nabla g$$

- In general these vectors are not unitary nor orthogonal
- We can also find $\mathbf{n}_f, \mathbf{n}_g$ looking for two independent vectors $\perp$ to $\mathbf{t}$

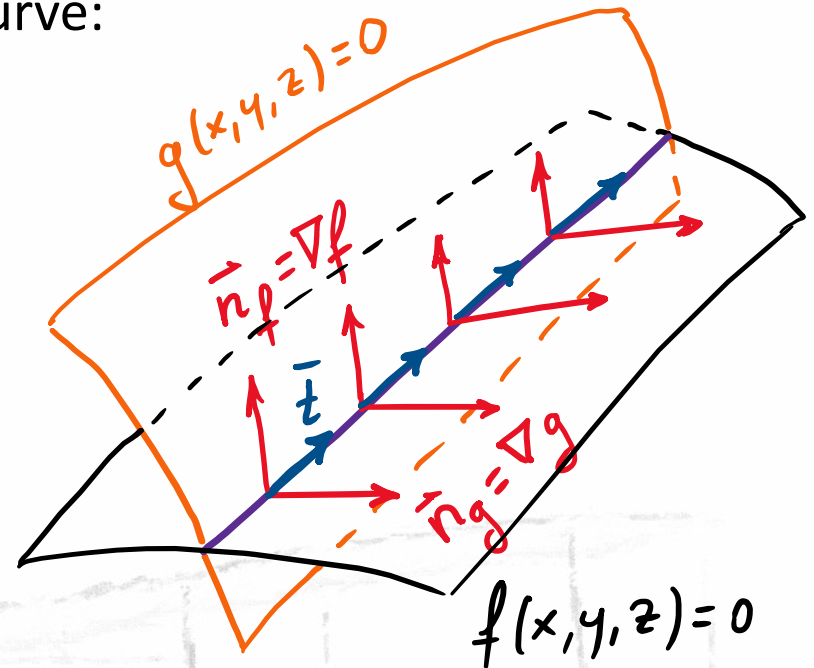
- We can obtain $\mathbf{t}$ by differentiation in the parametric expressions of the curve:

$$\begin{cases} x = x(\alpha) \\ y = y(\alpha) \\ z = z(\alpha) \end{cases} \Rightarrow \mathbf{t} = \frac{\partial x}{\partial \alpha} \mathbf{i} + \frac{\partial y}{\partial \alpha} \mathbf{j} + \frac{\partial z}{\partial \alpha} \mathbf{k}$$

- Observe that if we already have $\mathbf{n}_f, \mathbf{n}_g$, it is easy to find a valid tangent vector $\mathbf{t}$ by doing:

$$\mathbf{t} = \mathbf{n}_f \times \mathbf{n}_g$$

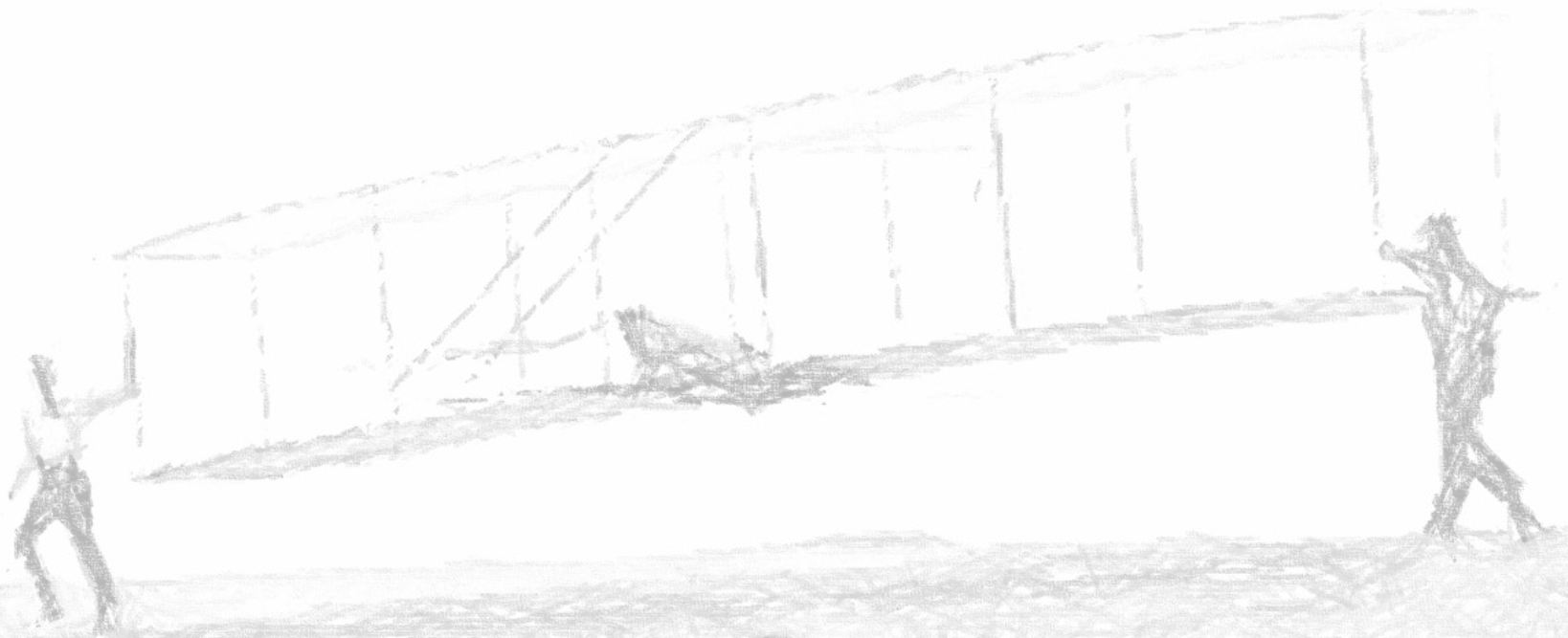
- $B_P = \{\mathbf{t}, \mathbf{n}_f, \mathbf{n}_g\}$ is a local vector basis at point P on the curve. Observe that this basis is generally not unitary, nor orthogonal, nor right-handed.



Example

- Find a tangent vector and two normal vectors for this curve:

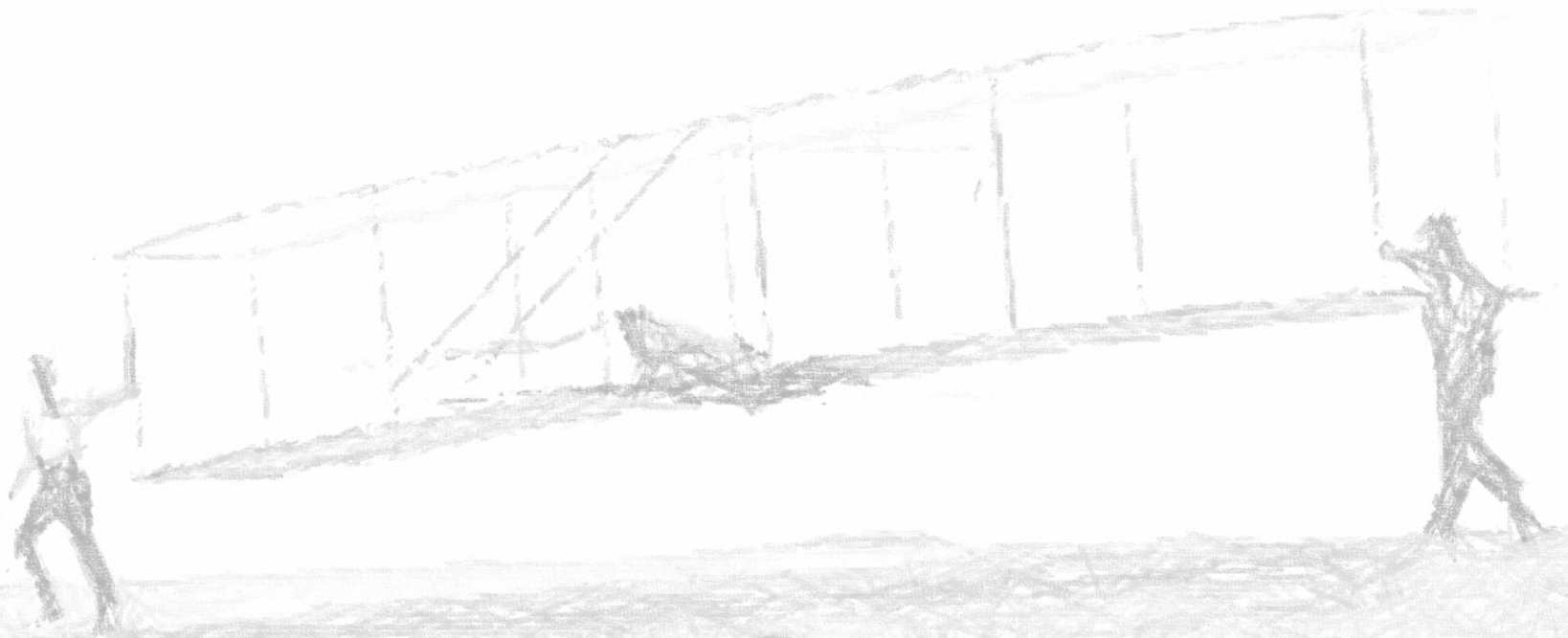
$$\begin{cases} x = a \cos \alpha \\ y = b \sin \alpha \\ z = 0 \end{cases}$$



Example

- Find two tangent vectors and a normal vector for this surface:

$$x^2 + y^2 = z; \quad z > 0$$



Constraint forces (aka reaction forces)

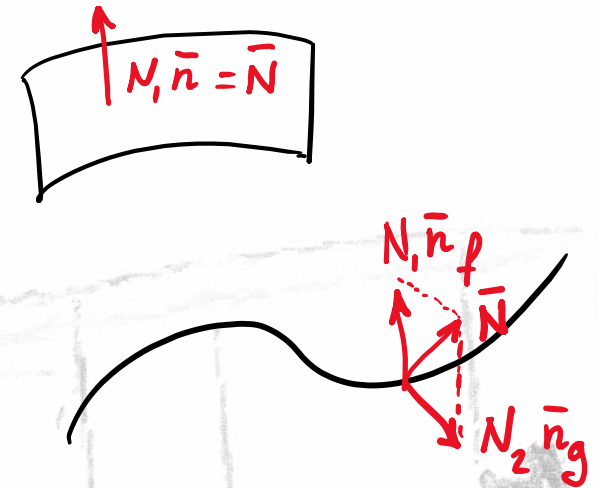
- The physical mechanism that ensures that a constraint is met is a *constraint (or reaction) force, $\mathbf{N}$* .
 - This constraint/reaction force is always in the normal subspace V_N of the constraint. In other words, $\mathbf{N} \in V_N$
 - The components of $\mathbf{N}$ are *unknown a priori*, and must be computed as part of the solution of the problem. We introduce new unknowns as follows:
 - $\mathbf{N} = N_1 \mathbf{n}$ in the case of a surface
($\dim V_N = 1$, so one unknown component)
 - $\mathbf{N} = N_1 \mathbf{n}_f + N_2 \mathbf{n}_g$ in the case of a curve
($\dim V_N = 2$, so two unknown component)

Applied forces: known a priori.

Constraint forces: components not known a priori

- To solve the problem we note:
 - We have 3 scalar equations of motion
 - In the case of a curve, we have 1 CDOF and 2 unknown reactions
 - In the case of a surface, we have 2 CDOFs and 1 unknown reaction

3 equations for 3 unknowns: the problem is closed

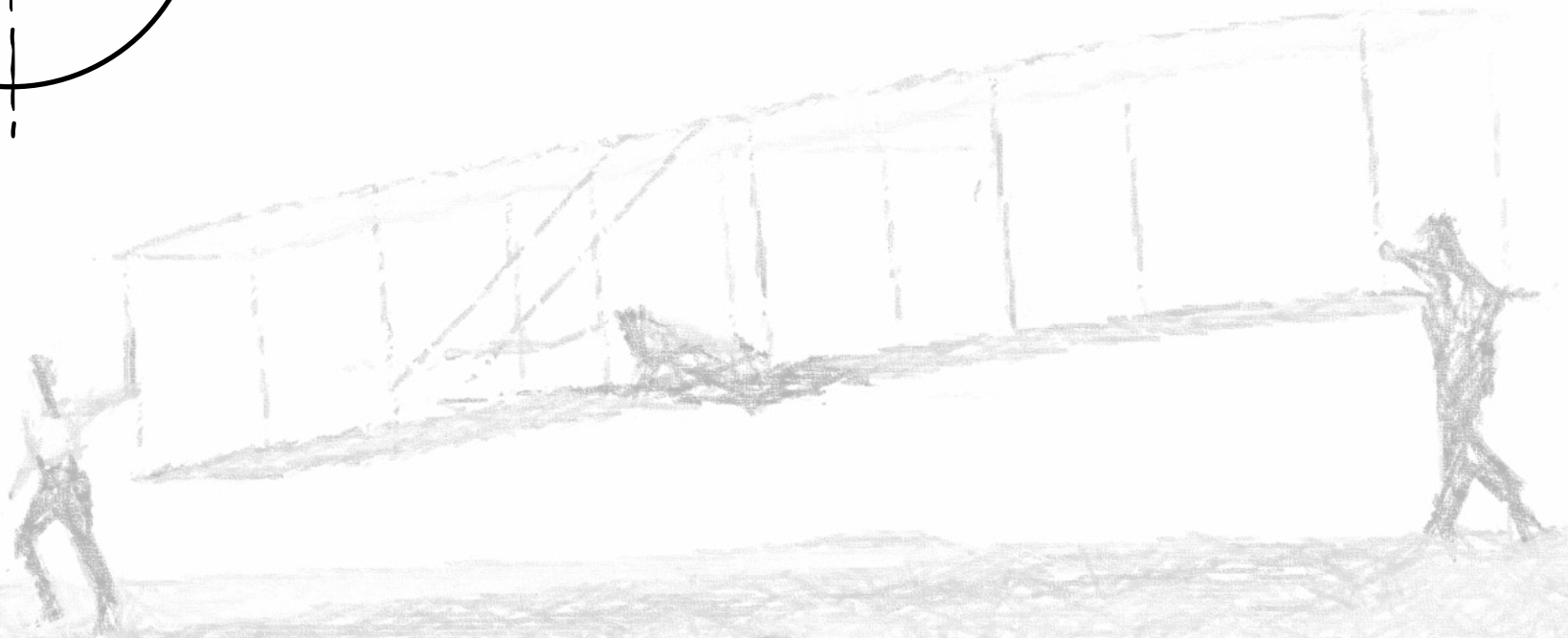
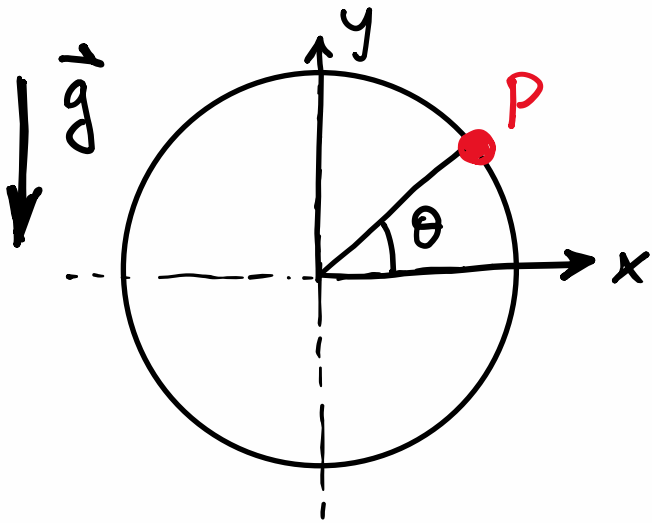


Constraint forces (aka reaction forces)

- We can solve for the coordinates as a function of time first, if we can isolate an equal number of equations of motion that are *independent of the reactions*
 - This can be done by projecting $m^P \mathbf{a}_0^P = \mathbf{F}$ along...
 - $\mathbf{t}$ (curve case, 1 equation needed)
 - $\mathbf{t}_1, \mathbf{t}_2$ (surface case, 2 equation needed)
 - Equivalently, you can work out linear combinations of the equations of motion projected along any three arbitrary independent vectors, until you get expressions where the unknown reactions do not appear
- The remaining equations contain $N_1, N_2, \dots$ and can be used afterwards to determine the reactions.
 - These equations can be obtained projecting $m^P \mathbf{a}_0^P = \mathbf{F}$ along...
 - $\mathbf{n}$ (surface case, 1 equation needed)
 - $\mathbf{n}_f, \mathbf{n}_g$ (curve case, 2 equation needed)
 - Or, again, by linear combination of the equations of motion projected along any three independent directions

Example

- A heavy particle P is constrained to move along a vertical circle of radius a . Get the equations of motion for the angle θ first, and then the equation that yields the magnitude of the constraint force N :



Power and work of a constraint force

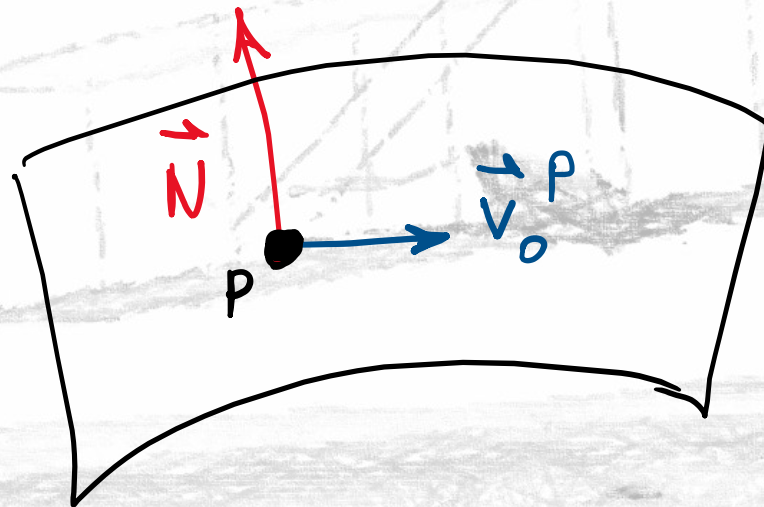
- The power of a constraint force $\mathbf{N}$ is equal to

$$\dot{W}_0 = \mathbf{N} \cdot \mathbf{v}_0^P$$

- $\mathbf{N} \in V_N$ always. For a stationary constraint (i.e., the constraint equation is independent of time t), the velocity vector is always contained in the tangent subspace, $\mathbf{v}_0^P \in V_T$
 - Hence, $\mathbf{N}$ and $\mathbf{v}_0^P$ are always orthogonal, and $\dot{W}_0 = 0$ (and $W_0 = 0$)

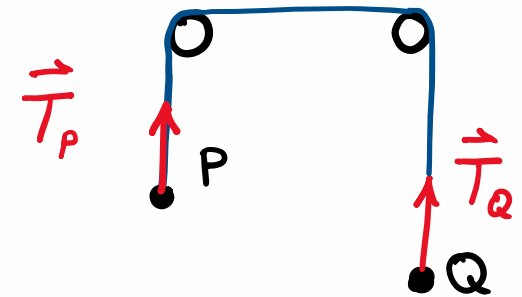
The constraint force of a stationary constraint never exerts power / work on the particle

- This is important: the existence of constraint forces does not decide the conservation of mechanical energy, as long as the constraint is stationary.



Constraints affecting two or more particles

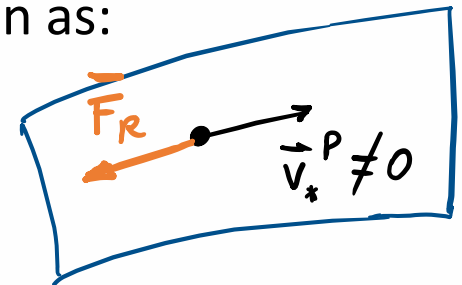
- In systems of particles, it is possible to have constraints that restrict the combined motion of two or more particles
 - E.g.: two particles connected by a massless, inextensible string, passing by a set of pulleys
- The treatment of the kinematics of the problem is the same: we describe the position of all particles with a minimal set of coordinates, so that the constraint equation is already “dealt with”
- The constraint forces on each of the affected particles are related
 - In the string example, the magnitude of the tension is equal on each particle
- The constraint force may exert work on each individual particle. However, if the constraint does not depend explicitly on time, the *total* mechanical energy of the *system* is conserved
 - In this example, the work exerted on the particle at one end of the string is equal and opposite to the work exerted on the other; the constraint only *transfers* mechanical energy between the particles



Friction

- When surfaces and curves are not *smooth*, they may cause some *friction* on the particle, $\mathbf{F}_R$
- When the particle is moving relative to the object ($\mathbf{v}_*^P \neq \mathbf{0}$), friction is an *applied force*: we model its magnitude and direction as:

$$\mathbf{F}_R = -\mu N \frac{\mathbf{v}_*^P}{v_*^P}$$



- μ is the friction coefficient; $\mathbf{v}_*^P$ the relative velocity
 - N is the magnitude of the normal constraint force
 - The friction force in this case always points in the direction opposite to the velocity. Obviously, it exerts (negative) power / work.
-
- When the particle is *not* moving relative to the constraint ($\mathbf{v}_*^P = \mathbf{0}$), friction acts like an additional *constraint force*. It is a vector contained in the tangent subspace of the surface/curve, $\mathbf{F}_R \in V_T$, but we do not know its components in the tangent basis:

When friction behaves as a constraint force, it introduces additional unknowns in the problem.

 - $\mathbf{F}_R = R_1 \mathbf{t}$, in the case of a curve
 - $\mathbf{F}_R = R_1 \mathbf{t}_\alpha + R_2 \mathbf{t}_\beta$, in the case of a surface
 - Friction as a constraint force does not exert work, because $\mathbf{v}_*^P = \mathbf{0}$

Friction

- To solve problems with friction with finite μ , we must study two separate formulations of the problem, and switch between them as needed:

- In the “constraint force” case, we work with the additional constraint $\mathbf{v}_*^P = \mathbf{0}$, as long as its magnitude satisfies:

$$F_R \leq \mu N$$

- If the reaction force needed to keep the particle at rest relative to the constraint were to become $F_R > \mu N$, then the particle will begin to move immediately, and we must switch to the “applied force” case, with

$$\mathbf{F}_R = -\mu N \frac{\mathbf{v}_*^P}{v_*^P}$$

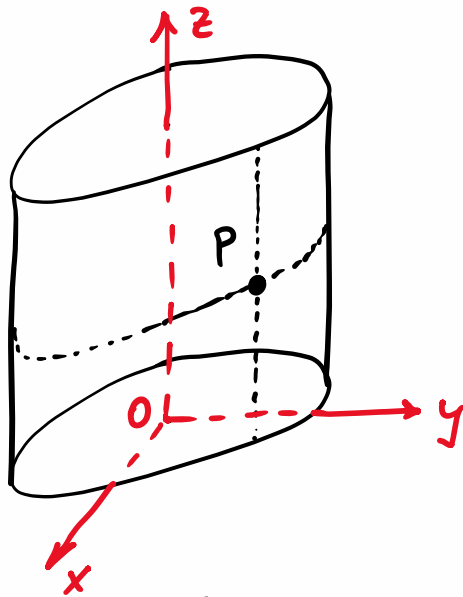
- The particle will move with respect to the constraint until $\mathbf{v}_*^P = \mathbf{0}$ again. At this point, we must switch back to the “constraint force” case. Etc.

Condition to switch from constraint force to applied force: $F_R = \mu N$
Condition to switch from applied force to constraint force: $\mathbf{v}_*^P = \mathbf{0}$

- There are other friction models besides the one presented here (Coulomb friction model), but we will not cover them in this course

Example

- A particle P (mass m) moves over the surface of a *rough* circular cylinder of radius a with friction coefficient μ . Write down the equations of motion in the “applied force” case and in the “constraint force” case.



Cylindrical coord:

$$\vec{r}_o^P = a \vec{e}_R + z \vec{e}_z$$

$$\vec{v}_o^P = a \dot{\theta} \vec{e}_\theta + \dot{z} \vec{e}_z$$

$$\vec{a}_o^P = -a \dot{\theta}^2 \vec{e}_R + a \ddot{\theta} \vec{e}_\theta + \ddot{z} \vec{e}_z$$

$$\vec{N} = N \vec{e}_R$$

Applied force case:

$$m \vec{a}_o^P = N \vec{e}_R - \mu N \frac{a \dot{\theta} \vec{e}_\theta + \dot{z} \vec{e}_z}{\sqrt{a^2 \dot{\theta}^2 + \dot{z}^2}}$$

Condition:

$$\sqrt{a^2 \dot{\theta}^2 + \dot{z}^2} > 0$$

Reaction force case:

$$0 = N \vec{e}_R + R_1 \vec{e}_\theta + R_2 \vec{e}_z$$

Condition:

$$\sqrt{R_1^2 + R_2^2} \leq \mu N$$

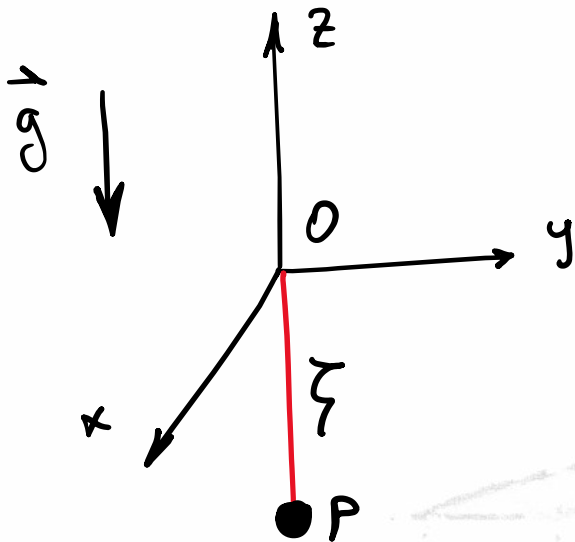
Advanced constraints

- There are two basic types of constraints:
 - If the constraint can be expressed in terms of the coordinates of a coordinate system, e.g. x, y, z , it is considered *integrable* or *holonomic*
 - If its expression necessarily involves the derivatives of the coordinates, $\dot{x}, \dot{y}, \dot{z}$, and it cannot be integrated, it is considered *non-integrable* or *non-holonomic*
 - Non-integrable constraints cannot be used to reduce the number of necessary constraints. We must include the constraint equation in the formulation of the problem
- Regarding its time dependency, a constraint can also be:
 - *Stationary*, if it does not depend explicitly on time
 - *Time-dependent* or *rheonomic*, otherwise
 - Time-dependent constraints may exert work on the particle
 - Any time-dependency forces us to add t (or a related variable) as an extra coordinate of the problem (+1 CDOF)
- If it is expressed in terms of equations or inequalities:
 - Bilateral. If the constraint is a surface, the particle may not leave it on either side
 - Unilateral (inequality). If the constraint is a surface, the particle may take-off from it on one side. E.g. An airplane on runway can take off to the sky, but may not dig into the ground
 - Unilateral constraints force us to consider two problems: when the constraint is active, and when it is not. We must switch from one problem to the other when the particle takes off ($N = 0$) or when it attaches back to the constraint

A simple constraint is
an integrable,
stationary, bilateral
constraint

Examples

- A heavy particle P of mass m is attached to a massless, inextensible string that is being collected into a spool of negligible radius located at O, at a rate c (length per unit time). The particle can only move along the vertical axis. Initially, the extended length of string is L . Discuss this constraint.



$$\vec{r}_O^P = -\zeta \vec{k}$$

$$\vec{v}_O^P = c \vec{k}$$

$$\vec{a}_O^P = \vec{0}$$

$$\begin{cases} \dot{\zeta} = -c \\ \zeta(0) = L \end{cases}$$

$$\vec{T} = T \vec{k}$$

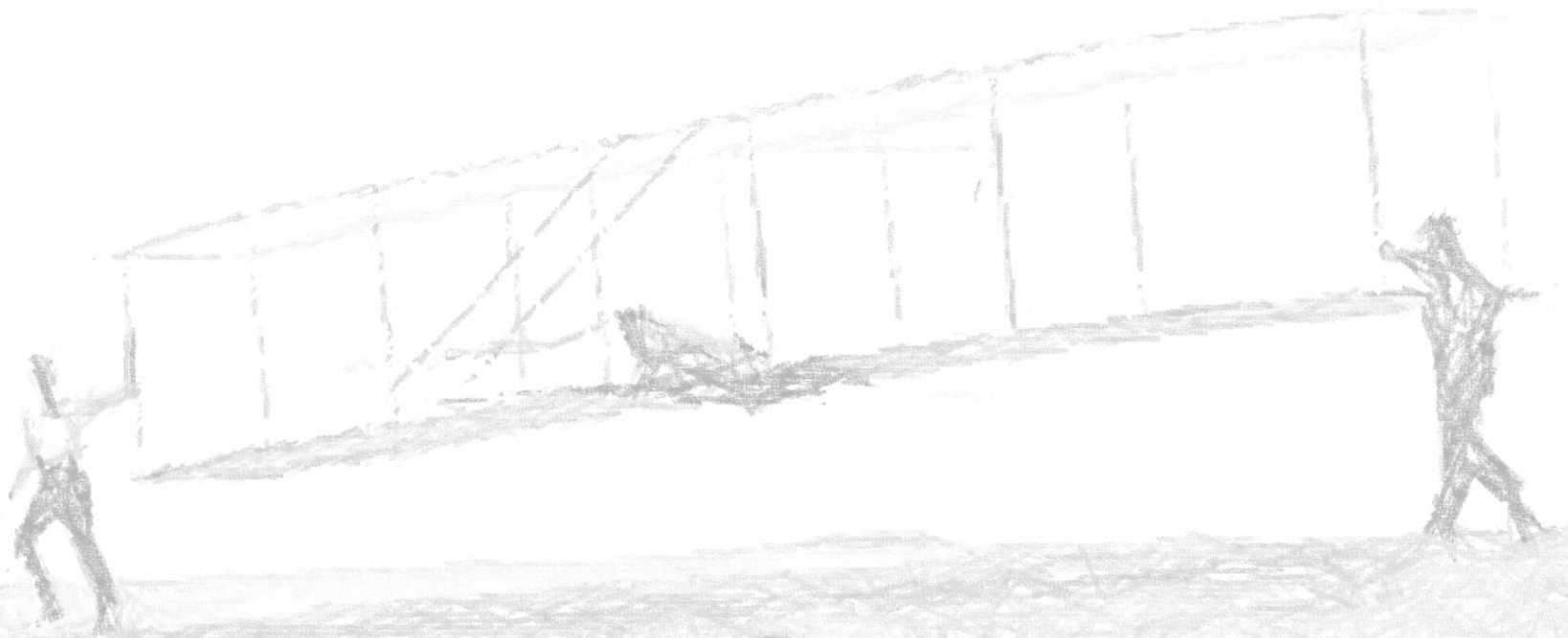
$$\vec{W} = -mg \vec{k}$$

$$m\vec{a}_O^P = -mg \vec{k} + T \vec{k} \rightarrow T = mg$$

$$\dot{W}_O = T \vec{k} \cdot c \vec{k} = cT \neq 0$$

Examples

- Consider an ideal airplane on the runway, in a stationary atmosphere. The airplane is heavy and has a mass m . The lift force is $L = \rho v^2 SC_L/2$. There is no drag. At what speed does the airplane lift off the ground?



Thank you!

